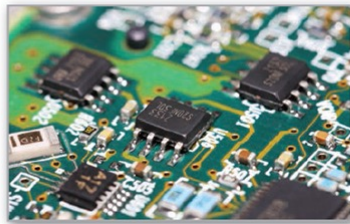
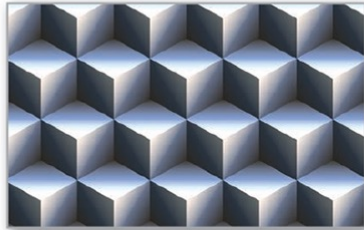


6 Linear Transformations



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1

6.1

Introduction to Linear Transformations

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2

Objectives

- Find the image and preimage of a function.
- Show that a function is a linear transformation, and find a linear transformation.

3

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Images And Preimages of Functions

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Images and Preimages of Functions (1 of 2)

In this section, we will learn about functions that **map** a vector space V into a vector space W . This type of function is denoted by

$$T: V \rightarrow W.$$

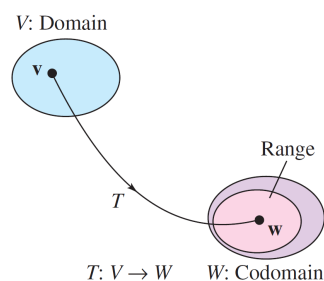
The standard function terminology is used for such functions. For instance, V is the **domain** of T , and W is the **codomain** of T .

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Images and Preimages of Functions (2 of 2)

If $\mathbf{v}$ is in V and $\mathbf{w}$ is in W such that $T(\mathbf{v}) = \mathbf{w}$, then $\mathbf{w}$ is the **image** of $\mathbf{v}$ under T . The set of all images of vectors in V is the **range** of T , and the set of all $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$ is the **preimage** of $\mathbf{w}$. (See below.)



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Example 1 – A function from R^2 into R^2

For any vector $\mathbf{v} = (v_1, v_2)$ in R^2 , define $T: R^2 \rightarrow R^2$ by

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2).$$

- a. Find the image of $\mathbf{v} = (-1, 2)$.
- b. Find the image of $\mathbf{v} = (0, 0)$.
- c. Find the preimage of $\mathbf{w} = (-1, 11)$.

Solution:

- a. For $\mathbf{v} = (-1, 2)$, you have

$$T(-1, 2) = (-1 - 2, -1 + 2(2)) = (-3, 3).$$

7

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Example 1 – Solution

- b. If $\mathbf{v} = (0, 0)$, then

$$T(0, 0) = (0 - 0, 0 + 2(0)) = (0, 0).$$

- c. If $T(\mathbf{v}) = (v_1 - v_2, v_1 + 2v_2) = (-1, 11)$, then

$$v_1 - v_2 = -1$$

$$v_1 + 2v_2 = 11.$$

This system of equations has the unique solution $v_1 = 3$ and $v_2 = 4$. So, the preimage of $(-1, 11)$ is the set in R^2 consisting of the single vector $(3, 4)$.

8

8

Linear Transformations

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9

Linear Transformations (1 of 4)

Definition of a Linear Transformation

Let V and W be vector spaces. The function

$$T: V \rightarrow W.$$

is a **linear transformation** of V into W when the two properties below are true for all $\mathbf{u}$ and $\mathbf{v}$ in V and for any scalar c .

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$

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Example 2 – Verifying a Linear Transformation from R^2 into R^2

Show that the function in Example 1 is a linear transformation from R^2 into R^2 .

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

Solution:

Let $\mathbf{v} = (v_1, v_2)$ and $\mathbf{u} = (u_1, u_2)$ be vectors in R^2 and let c be any real number.

11

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Example 2 – Solution (1 of 2)

Then, using the properties of vector addition and scalar multiplication, you have the two statements below.

1. $\mathbf{u} + \mathbf{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$, so you have

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(u_1 + v_1, u_2 + v_2) \\ &= ((u_1 + v_1) - (u_2 + v_2), (u_1 + v_1) + 2(u_2 + v_2)) \\ &= ((u_1 - u_2) + (v_1 - v_2), (u_1 + 2u_2) + (v_1 + 2v_2)) \\ &= (u_1 - u_2, u_1 + 2u_2) + (v_1 - v_2, v_1 + 2v_2) \\ &= T(\mathbf{u}) + T(\mathbf{v}). \end{aligned}$$

12

12

Example 2 – Solution (2 of 2)

2. $c\mathbf{u} = c(u_1, u_2) = (cu_1, cu_2)$, so you have

$$\begin{aligned} T(c\mathbf{u}) &= T(cu_1, cu_2) \\ &= (cu_1 - cu_2, cu_1 + 2cu_2) \\ &= c(u_1 - u_2, u_1 + 2u_2) \\ &= cT(\mathbf{u}). \end{aligned}$$

So, T is a linear transformation.

13

13

Linear Transformations (2 of 4)

Theorem 6.1 Properties of Linear Transformations

Let T be a linear transformation from V into W , where $\mathbf{u}$ and $\mathbf{v}$ are in V . Then the properties listed below are true.

1. $T(\mathbf{0}) = \mathbf{0}$
2. $T(-\mathbf{v}) = -T(\mathbf{v})$
3. $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$
4. If $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$, then

$$\begin{aligned} T(\mathbf{v}) &= T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n) \\ &= c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \dots + c_nT(\mathbf{v}_n). \end{aligned}$$

14

14

Example 4 – Linear Transformations and Bases

Let $T: R^3 \rightarrow R^3$ be a linear transformation such that

$$T(1, 0, 0) = (2, -1, 4)$$

$$T(0, 1, 0) = (1, 5, -2)$$

$$T(0, 0, 1) = (0, 3, 1).$$

Find $T(2, 3, -2)$.

15

15

Example 4 – Solution

$(2, 3, -2) = 2(1, 0, 0) + 3(0, 1, 0) - 2(0, 0, 1)$, so use Property 4 of Theorem 6.1 to write

$$\begin{aligned} T(2, 3, -2) &= 2T(1, 0, 0) + 3T(0, 1, 0) - 2T(0, 0, 1) \\ &= 2(2, -1, 4) + 3(1, 5, -2) - 2(0, 3, 1) \\ &= (7, 7, 0). \end{aligned}$$

16

16

Example 5 – A Linear Transformation Defined by a Matrix

Define the function $T: R^2 \rightarrow R^3$ as

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

- Find $T(\mathbf{v})$ when $\mathbf{v} = (2, -1)$.
- Show that T is a linear transformation from R^2 into R^3 .

17

17

Example 5 – Solution (1 of 2)

- $\mathbf{v} = (2, -1)$, so you have

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}$$



Vector
in R^2



Vector
in R^3

which means that $T(2, -1) = (6, 3, 0)$.

18

18

Example 5 – Solution (2 of 2)

- b.** Begin by observing that T maps a vector in R^2 to a vector in R^3 . To show that T is a linear transformation. For any vectors $\mathbf{u}$ and $\mathbf{v}$ in R^2 , the distributive property of matrix multiplication over addition produces

$$T(\mathbf{u} + \mathbf{v}) = A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = T(\mathbf{u}) + T(\mathbf{v}).$$

Similarly, for any vector $\mathbf{u}$ in R^2 and any scalar c , the commutative property of scalar multiplication with matrix multiplication produces

$$T(c\mathbf{u}) = A(c\mathbf{u}) = c(A\mathbf{u}) = cT(\mathbf{u}).$$

19

19

Linear Transformations (3 of 4)

Theorem 6.2 Linear Transformation Given by a Matrix

Let A be an $m \times n$ matrix. The function T defined by

$$T(\mathbf{v}) = A\mathbf{v}$$

is a linear transformation from R^n into R^m . In order to conform to matrix multiplication with an $m \times n$ matrix, $n \times 1$ matrices represent the vectors in R^n and $m \times 1$ matrices represent the vectors in R^m .

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20

Example 6 – Linear Transformations Given by Matrices

Consider the linear transformation $T: R^n \rightarrow R^m$ defined by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$. Find the dimensions of R^n and R^m for the linear transformation represented by each matrix.

$$\begin{array}{ll} \text{a. } A = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 3 & 0 \\ 4 & 2 & 1 \end{bmatrix} & \text{b. } A = \begin{bmatrix} 2 & -3 \\ -5 & 0 \\ 0 & -2 \end{bmatrix} \\ \text{c. } A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 3 & 1 & 0 & 0 \end{bmatrix} & \end{array}$$

21

21

Example 6 – Solution (1 of 2)

- a. The size of this matrix is 3×3 , so it defines a linear transformation from R^3 into R^3 .

$$A\mathbf{v} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 3 & 0 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$



Vector
in R^3



Vector
in R^3

22

22

Example 6 – Solution (2 of 2)

- b.** The size of this matrix is 3×2 , so it defines a linear transformation from R^2 into R^3 .
- c.** The size of this matrix is 2×4 , so it defines a linear transformation from R^4 into R^2 .

23

23

Example 10 – The Differential Operator (Calculus)

Let $C'[a, b]$ be the set of all functions whose derivatives are continuous on $[a, b]$. Show that the differential operator D_x defines a linear transformation from $C'[a, b]$ into $C[a, b]$.

Solution:

Using operator notation, you can write

$$D_x(f) = \frac{d}{dx}[f]$$

where f is in $C'[a, b]$. To show that D_x is a linear transformation, you must use calculus.

24

24

Example 10 – Solution (1 of 2)

Specifically, the derivative of the sum of two differentiable functions is equal to the sum of their derivatives, so you have

$$D_x(f + g) = \frac{d}{dx}[f + g] = \frac{d}{dx}[f] + \frac{d}{dx}[g] = D_x(f) + D_x(g)$$

where g is also in $C'[a, b]$. Similarly, the derivative of a scalar multiple cf of a differentiable function is equal to the scalar multiple of the derivative, so you have

$$D_x(cf) = \frac{d}{dx}[cf] = c\left(\frac{d}{dx}[f]\right) = cD_x(f).$$

25

25

Example 10 – Solution (2 of 2)

The sum of two continuous functions is continuous, and the scalar multiple of a continuous function is continuous, so D_x is a linear transformation from $C'[a, b]$ into $C[a, b]$.

26

26

Linear Transformations (4 of 4)

The linear transformation D_x is called the **differential operator**. For polynomials, the differential operator is a linear transformation from P_n into P_{n-1} because the derivative of a polynomial function of degree $n \geq 1$ is a polynomial function of degree $n - 1$. That is,

$$D_x(a_0 + a_1x + \cdots + a_nx^n) = a_1 + \cdots + na_nx^{n-1}.$$

27

27

Example 11 – The Definite Integral as a Linear Transformation (Calculus)

Consider $T: P \rightarrow R$ defined by

$$T(p) = \int_a^b p(x) \, dx$$

where p is a polynomial function. Show that T is a linear transformation from P , the vector space of polynomial functions, into R , the vector space of real numbers.

28

28

Example 11 – Solution

Using properties of definite integrals, you can write

$$T(p + q) = \int_a^b [p(x) + q(x)] dx = \int_a^b p(x) dx + \int_a^b q(x) dx = T(p) + T(q)$$

where q is a polynomial function, and

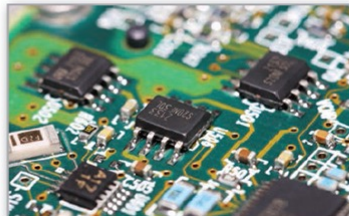
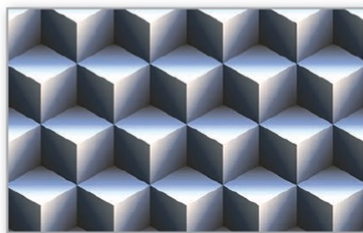
$$T(cp) = \int_a^b [cp(x)] dx = c \int_a^b p(x) dx = cT(p)$$

where c is a scalar. So, T is a linear transformation.

29

29

6 Linear Transformations



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30

6.2

The Kernel and Range of a Linear Transformation

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31

Objectives

- Find the kernel of a linear transformation.
- Find a basis for the range, the rank, and the nullity of a linear transformation.
- Determine whether a linear transformation is one-to-one or onto.
- Determine whether vector spaces are isomorphic.

32

32

The Kernel of a Linear Transformation

33

33

The Kernel of a Linear Transformation (1 of 3)

Definition of Kernel of a Linear Transformation

Let $T: V \rightarrow W$ be a linear transformation.

Then the set of all vectors $\mathbf{v}$ in V that satisfy $T(\mathbf{v}) = \mathbf{0}$ is the **kernel** of T and is denoted by $\ker(T)$.

34

34

Example 3 – Finding the Kernel of a Linear Transformation

Find the kernel of the projection $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ represented by $T(x, y, z) = (x, y, 0)$.

Solution:

This linear transformation projects the vector (x, y, z) in $\mathbb{R}^3$ to the vector $(x, y, 0)$ in the xy -plane.

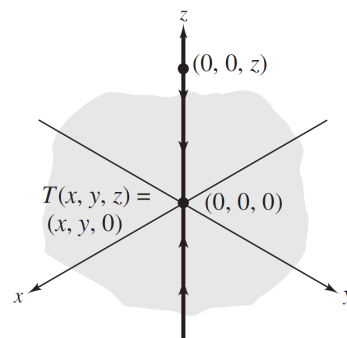
The kernel consists of all vectors lying on the z -axis.

35

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Example 3 – Solution

That is, $\ker(T) = \{(0, 0, z) : z \text{ is a real number}\}$. See Figure 6.1.



The kernel of T is the set of all vectors on the z -axis.

Figure 6.1

36

36

The Kernel of a Linear Transformation (2 of 3)

Theorem 6.3 The Kernel is a subspace of V

The kernel of a linear transformation $T: V \rightarrow W$ is a subspace of the domain V .

37

37

Example 6 – Finding a Basis for the Kernel

Define $T: R^5 \rightarrow R^4$ by $T(\mathbf{x}) = A\mathbf{x}$, where $\mathbf{x}$ is in R^5 and

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix}.$$

Find a basis for $\ker(T)$ as a subspace of R^5 .

38

38

Example 6 – Solution (1 of 2)

Write the augmented matrix $[A \ 0]$ in reduced row-echelon form as shown below.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{aligned} x_1 &= -2x_3 + x_5 \\ x_2 &= x_3 + 2x_5 \\ x_4 &= -4x_5 \end{aligned}$$

39

39

Example 6 – Solution (2 of 2)

Letting $x_3 = s$ and $x_5 = t$, you have

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s + t \\ s + 2t \\ s + 0t \\ 0s - 4t \\ 0s + t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \\ -4 \\ 1 \end{bmatrix}.$$

So a basis for the kernel of T is

$$B = \{(-2, 1, 1, 0, 0), (1, 2, 0, -4, 1)\}.$$

40

40

The Kernel of a Linear Transformation (3 of 3)

Theorem 6.3 Corollary

Let $T: R^n \rightarrow R^m$ be the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.

Then the kernel of T is equal to the solution space of $A\mathbf{x} = \mathbf{0}$.

41

41

The Range of a Linear Transformation

42

42

The Range of a Linear Transformation (1 of 4)

The kernel is one of two critical subspaces associated with a linear transformation. The other is the range of T , denoted by $\text{range}(T)$. The range of $T: V \rightarrow W$ is the set of all vectors $\mathbf{w}$ in W that are images of vectors in V . That is,

$$\text{range}(T) = \{T(\mathbf{v}) : \mathbf{v} \text{ is in } V\}.$$

Theorem 6.4 The Range of T Is a Subspace of W

The range of a linear transformation $T: V \rightarrow W$ is a subspace of W .

43

43

The Range of a Linear Transformation (2 of 4)

Theorem 6.4 Corollary

Let $T: R^n \rightarrow R^m$ be the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.

Then the column space of A is equal to the range of T .

44

44

Example 7 – Finding a Basis for the range of a linear Transformation

For the linear transformation $R^5 \rightarrow R^4$ from previous example, find a basis for the range of T .

Solution:

Use the reduced row-echelon form of A from previous example.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

45

45

Example 7 – Solution

The leading 1's appear in columns 1, 2, and 4 of the reduced matrix on the right, so the corresponding column vectors of A form a basis for the column space of A .

A basis for the range of T is

$$B = \{(1, 2, -1, 0), (2, 1, 0, 0), (1, 1, 0, 2)\}.$$

46

46

The Range of a Linear Transformation (3 of 4)

Definition of rank and nullity of a linear Transformation

Let $T: V \rightarrow W$ be a linear transformation. The dimension of the kernel of T is called the **nullity** of T and is denoted by **nullity**(T).

The dimension of the range of T is called the **rank** of T and is denoted by **rank**(T).

47

47

The Range of a Linear Transformation (4 of 4)

Theorem 6.5 Sum of rank and nullity

Let $T: V \rightarrow W$ be a linear transformation from an n -dimensional vector space V into a vector space W . Then the sum of the dimensions of the range and kernel is equal to the dimension of the domain. That is,

$$\text{rank}(T) + \text{nullity}(T) = n$$

or

$$\dim(\text{range}) + \dim(\text{kernel}) = \dim(\text{domain}).$$

48

48

Example 9 – Finding the rank and nullity of a linear Transformation

Let $T: R^5 \rightarrow R^7$ be a linear transformation.

- Find the dimension of the kernel of T when the dimension of the range is 2.
- Find the rank of T when the nullity of T is 4.
- Find the rank of T when $\ker(T) = \{0\}$.

Solution:

- By Theorem 6.5, with $n = 5$, you have

$$\dim(\text{kernel}) = n - \dim(\text{range}) = 5 - 2 = 3.$$

49

49

Example 9 – Solution

- Again by Theorem 6.5, you have

$$\text{rank}(T) = n - \text{nullity}(T) = 5 - 4 = 1.$$

- In this case, the nullity of T is 0. So

$$\text{rank}(T) = n - \text{nullity}(T) = 5 - 0 = 5.$$

50

50

One-to-One and Onto Linear Transformations

51

51

One-to-One and Onto Linear Transformations (1 of 3)

Theorem 6.6 One-to-One Linear Transformations

Let $T: V \rightarrow W$ be a linear transformation. Then T is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$.

52

52

Example 10 – One-to-One and Not One-to-One Linear Transformations

- a. The linear transformation $T: M_{m,n} \rightarrow M_{n,m}$ represented by $T(A) = A^T$ is one-to-one because its kernel consists of only the $m \times n$ zero matrix.
- b. The zero transformation $T: R^3 \rightarrow R^3$ is not one-to-one because its kernel is all R^3 .

53

53

One-to-One and Onto Linear Transformations (2 of 3)

Theorem 6.7 Onto Linear Transformations

Let $T: V \rightarrow W$ be a linear transformation, where W is finite dimensional.

Then T is onto if and only if the rank of T is equal to the dimension of W .

54

54

One-to-One and Onto Linear Transformations (3 of 3)

Theorem 6.8 One-to-One and Onto Linear Transformations

Let $T: V \rightarrow W$ be a linear transformation with vector spaces V and W , both of dimension n . Then T is one-to-one if and only if it is onto.

55

55

Example 11 – Summarizing Several Results

Consider the linear transformation $T: R^n \rightarrow R^m$ represented by $T(\mathbf{x}) = A\mathbf{x}$. Find the nullity and rank of T , and determine whether T is one-to-one, onto, or neither.

a. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

56

56

Example 11 – Solution

Note that each matrix is already in row-echelon form, so its rank can be determined by inspection.

$T: R^n \rightarrow R^m$	Dim(domain)	Dim(range) Rank(T)	Dim(kernel) Nullity(T)	One-to-One	Onto
a. $T: R^3 \rightarrow R^3$	3	3	0	Yes	Yes
b. $T: R^2 \rightarrow R^3$	2	2	0	Yes	No
c. $T: R^3 \rightarrow R^2$	3	2	1	No	Yes
d. $T: R^3 \rightarrow R^3$	3	2	1	No	No

57

57

Isomorphisms of Vector Spaces

58

58

Isomorphisms of Vector Spaces (1 of 2)

Definition of Isomorphism

A linear transformation $T: V \rightarrow W$ that is one-to-one and onto is called an **isomorphism**.

Moreover, if V and W are vector spaces such that there exists an isomorphism from V to W , then V and W are **isomorphic** to each other.

59

59

Isomorphisms of Vector Spaces (2 of 2)

Theorem 6.9 Isomorphic Spaces and Dimension

Two finite-dimensional vector spaces V and W are isomorphic if and only if they are of the same dimension.

60

60

Example 12 – Isomorphic Vector Spaces

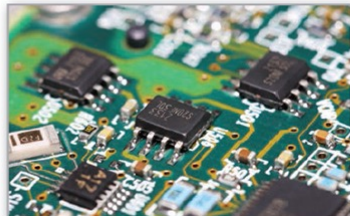
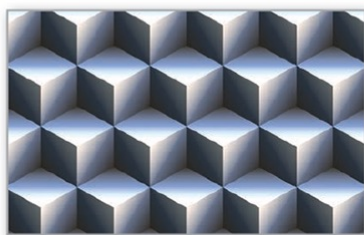
The vector spaces below are isomorphic to each other.

- a. R^4 = 4-space
- b. $M_{4,1}$ = space of all 4×1 matrices
- c. $M_{2,2}$ = space of all 2×2 matrices
- d. P_3 = space of all polynomials of degree 3 or less
- e. V = $\{(x_1, x_2, x_3, x_4, 0) : x_i \text{ is a real number}\}$ (subspace of R^5)

61

61

6 Linear Transformations



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62

6.3 Matrices for Linear Transformations

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63

Objectives

- Find the standard matrix for a linear transformation.
- Find the standard matrix for the composition of linear transformations and find the inverse of an invertible linear transformation.
- Find the matrix for a linear transformation relative to a nonstandard basis.

64

64

The Standard Matrix for a Linear Transformation

65

65

The Standard Matrix for a Linear Transformation (1 of 2)

Theorem 6.10 Standard Matrix for a Linear Transformation

Let $T: R^n \rightarrow R^m$ be a linear transformation such that, for the standard basis vectors $\mathbf{e}_i$ of R^n

$$T(\mathbf{e}_1) = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, T(\mathbf{e}_2) = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, T(\mathbf{e}_n) = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

66

66

The Standard Matrix for a Linear Transformation (2 of 2)

Theorem 6.10 Standard Matrix for a Linear Transformation

Then the $m \times n$ matrix whose n columns correspond to $T(\mathbf{e}_i)$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & \cdot & a_{2n} \\ \vdots & \vdots & & & & \vdots \\ a_{m1} & a_{m2} & \cdot & \cdot & \cdot & a_{mn} \end{bmatrix}$$

is such that $T(\mathbf{v}) = A\mathbf{v}$ for every $\mathbf{v}$ in R^n . A is called the **standard matrix** for T .

67

67

Example 1 – Finding the Standard Matrix for a Linear Transformation

Find the standard matrix for the linear transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - 2y, 2x + y)$.

Solution:

Begin by finding the images of $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$.

Vector Notation

Matrix Notation

$$T(\mathbf{e}_1) = T(1, 0, 0) = (1, 2)$$

$$T(\mathbf{e}_1) = T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$T(\mathbf{e}_2) = T(0, 1, 0) = (-2, 1)$$

$$T(\mathbf{e}_2) = T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$T(\mathbf{e}_3) = T(0, 0, 1) = (0, 0)$$

$$T(\mathbf{e}_3) = T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

68

68

Example 1 – Solution

By Theorem 6.10, the columns of A consist of $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, and $T(\mathbf{e}_3)$, and you have

$$A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ T(\mathbf{e}_3)] = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{bmatrix}.$$

69

69

Composition of Linear Transformations

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70

Composition of Linear Transformations (1 of 4)

The **composition**, T , of $T_1: R^n \rightarrow R^m$ with $T_2: R^m \rightarrow R^p$ is $T(\mathbf{v}) = T_2(T_1(\mathbf{v}))$

where $\mathbf{v}$ is a vector in R^n . This composition is denoted by $T = T_2 \circ T_1$.

71

71

Composition of Linear Transformations (2 of 4)

Theorem 6.11 Composition of Linear Transformations

Let $T_1: R^n \rightarrow R^m$ and $T_2: R^m \rightarrow R^p$ be linear transformations with standard matrices A_1 and A_2 , respectively.

The composition $T: R^n \rightarrow R^p$, defined by $T(\mathbf{v}) = T_2(T_1(\mathbf{v}))$, is a linear transformation. Moreover, the standard matrix A for T is the matrix product

$$A = A_2 A_1.$$

72

72

Example 3 – The Standard Matrix for a Composition

Let T_1 and T_2 be linear transformations from R^3 into R^3 such that $T_1(x, y, z) = (2x + y, 0, x + z)$ and $T_2(x, y, z) = (x - y, z, y)$.

Find the standard matrices for the compositions

$$T = T_2 \circ T_1 \text{ and } T' = T_1 \circ T_2.$$

Solution:

The standard matrices for T_1 and T_2 are

$$A_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ and } A_2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

73

73

Example 3 – Solution (1 of 2)

By Theorem 6.11, the standard matrix for T is

$$A = A_2 A_1$$

$$\begin{aligned} &= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

74

74

Example 3 – Solution (2 of 2)

And the standard matrix for T' is

$$\begin{aligned} A' &= A_1 A_2 \\ &= \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -2 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \end{aligned}$$

75

75

Composition of Linear Transformations (3 of 4)

Definition of Inverse Linear Transformation

If $T_1: R^n \rightarrow R^n$ and $T_2: R^n \rightarrow R^n$ are linear transformations such that for every $\mathbf{v}$ in R^n ,

$$T_2(T_1(\mathbf{v})) = \mathbf{v} \text{ and } T_1(T_2(\mathbf{v})) = \mathbf{v}$$

then T_2 is the **inverse** of T_1 , and T_1 is said to be **invertible**.

76

76

Composition of Linear Transformations (4 of 4)

Theorem 6.12 Existence of an Inverse Transformation

Let $T: R^n \rightarrow R^n$ be a linear transformation with standard matrix A . Then the conditions listed below are equivalent.

1. T is invertible.
2. T is an isomorphism.
3. A is invertible.

If T is invertible with standard matrix A , then the standard matrix for T^{-1} is A^{-1} .

77

77

Example 4 – Finding the Inverse of a Linear Transformation

Consider the linear transformation $T: R^3 \rightarrow R^3$ defined by
 $T(x_1, x_2, x_3) = (2x_1 + 3x_2 + x_3, 3x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + x_3)$.
 Show that T is invertible, and find its inverse.

Solution:

The standard matrix for T is

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}.$$

78

78

Example 4 – Solution (1 of 1)

Using the method inverse of a matrix or a graphing calculator, you can find that A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}.$$

So, T is invertible and the standard matrix for T^{-1} is A^{-1} .

79

79

Nonstandard Bases and General Vector Spaces

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80

Nonstandard Bases and General Vector Spaces (1 of 4)

To represent the linear transformation T , multiply A by a *coordinate matrix relative to B* to obtain a *coordinate matrix relative to B'* .

That is, $[T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B$. The matrix A is called the **matrix of T relative to the bases B and B'** .

81

81

Nonstandard Bases and General Vector Spaces (2 of 4)

Transformation Matrix for Nonstandard Bases

Let V and W be finite-dimensional vector spaces with bases B and B' , respectively, where

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}.$$

If $T: V \rightarrow W$ is a linear transformation such that

$$[T(\mathbf{v}_1)]_{B'} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, [T(\mathbf{v}_2)]_{B'} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, [T(\mathbf{v}_n)]_{B'} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

82

82

Nonstandard Bases and General Vector Spaces (3 of 4)

Then the $m \times n$ matrix whose n columns correspond to $[T(\mathbf{v}_i)]_{B'}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $[T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B$ for every $\mathbf{v}$ in V .

83

83

Example 5 – Finding a Matrix Relative to Nonstandard Bases

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2)$. Find the matrix for T relative to the bases

$$B = \{\overset{\mathbf{v}_1}{(1, 2)}, \overset{\mathbf{v}_2}{(-1, 1)}\} \quad \text{and} \quad B' = \{\overset{\mathbf{w}_1}{(1, 0)}, \overset{\mathbf{w}_2}{(0, 1)}\}.$$

Solution:

By the definition of T , you have

$$T(\mathbf{v}_1) = T(1, 2) = (3, 0) = 3\mathbf{w}_1 + 0\mathbf{w}_2$$

$$T(\mathbf{v}_2) = T(-1, 1) = (0, -3) = 0\mathbf{w}_1 - 3\mathbf{w}_2.$$

84

84

Example 5 – Solution

The coordinate matrices for $T(\mathbf{v}_1)$ and $T(\mathbf{v}_2)$ relative to B' are

$$[T(\mathbf{v}_1)]_{B'} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad \text{and} \quad [T(\mathbf{v}_2)]_{B'} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}.$$

Form the matrix for T relative to B and B' by using these coordinate matrices as columns to produce

$$A = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}.$$

85

85

Nonstandard Bases and General Vector Spaces (4 of 4)

For the special case where $V = W$ and $B = B'$, the matrix A is called the **matrix of T relative to the basis B** . In this case, the matrix of the identity transformation is simply I_n .

To see this, let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. The identity transformation maps each $\mathbf{v}_i$ to itself, so you have $[T(\mathbf{v}_1)]_B = [1 \ 0 \ \dots \ 0]^T, [T(\mathbf{v}_2)]_B = [0 \ 1 \ \dots \ 0]^T, \dots, [T(\mathbf{v}_n)]_B = [0 \ 0 \ \dots \ 1]^T$, and it follows that $A = I_n$.

86

86

Example 7 – A Matrix for the Differential Operator (Calculus)

Let $D_x: P_2 \rightarrow P_1$ be the differential operator that maps a polynomial p of degree 2 or less onto its derivative p' . Find the matrix for D_x using the bases

$$B = \{1, x, x^2\} \quad \text{and} \quad B' = \{1, x\}.$$

Solution:

The derivatives of the basis vectors are

$$D_x(1) = 0 = 0(1) + 0(x)$$

$$D_x(x) = 1 = 1(1) + 0(x)$$

$$D_x(x^2) = 2x = 0(1) + 2(x).$$

87

87

Example 7 – Solution (1 of 2)

So, the coordinate matrices relative to B' are

$$[D_x(1)]_{B'} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad [D_x(x)]_{B'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad [D_x(x^2)]_{B'} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

and the matrix for D_x is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

88

88

Example 7 – Solution (2 of 2)

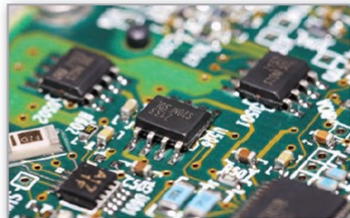
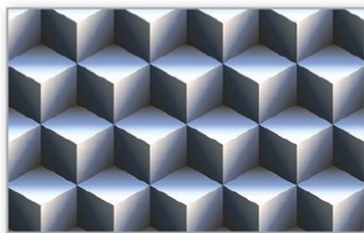
Note that this matrix *does* produce the derivative of a quadratic polynomial $p(x) = a + bx + cx^2$.

$$Ap = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix} \Rightarrow b + 2cx = D_x[a + bx + cx^2]$$

89

89

6 Linear Transformations



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90

6.4**Transition Matrices and Similarity**

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91

Objectives

- Find and use a matrix for a linear transformation.
- Show that two matrices are similar and use the properties of similar matrices.

92

92

The Matrix for a Linear Transformation

93

93

The Matrix for a Linear Transformation (1 of 1)

the matrix for a linear transformation $T: V \rightarrow V$ depends on the basis for V .

In other words, the matrix for T relative to a basis B is different from the matrix for T relative to another basis B' .

94

94

Example 1 – Finding a Matrix for a Linear Transformation

Find the matrix A' for $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where

$T(x_1, x_2) = (2x_1 - 2x_2, -x_1 + 3x_2)$, relative to the basis
 $B' = \{(1, 0), (1, 1)\}$.

Solution:

The standard matrix for T is $A = \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix}$.

95

95

Example 1 – Solution (1 of 2)

Furthermore, using the techniques of Section 4.7, the transition matrix from B' to the standard basis

$B = \{(1, 0), (0, 1)\}$ is

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

The inverse of this matrix is the transition matrix from B to B' ,

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

96

96

Example 1 – Solution (2 of 2)

So, the matrix for T relative to B' is

$$A' = P^{-1}AP = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}.$$

97

97

Similar Matrices

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98

Similar Matrices(1 of 2)

Two square matrices A and A' that are related by an equation $A' = P^{-1}AP$ are called **similar** matrices, as stated in the next definition.

Definition of similar Matrices

For square matrices A and A' of order n , A' is **similar** to A when there exists an invertible matrix P such that $A' = P^{-1}AP$

99

99

Similar Matrices(2 of 2)

If A' is similar to A , then it is also true that A is similar to A' , as stated in the next theorem. So, it makes sense to simply say that **A and A' are similar**.

Theorem 6.13 Properties of similar Matrices

Let A , B , and C be square matrices of order n . Then the properties below are true.

1. A is similar to A .
2. If A is similar to B , then B is similar to A .
3. If A is similar to B and B is similar to C , then A is similar to C .

100

100

Example 5 – A Comparison of Two Matrices for a Linear Transformation

Let

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

be the matrix for $T: R^3 \rightarrow R^3$ relative to the standard basis.
Find the matrix for T relative to the basis

$$B' = \{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}.$$

101

101

Example 5 – Solution (1 of 2)

The transition matrix from B' to the standard basis has columns consisting of the vectors in B' ,

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and it follows that

$$P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

102

102

Example 5 – Solution (2 of 2)

So, the matrix for T relative to B' is

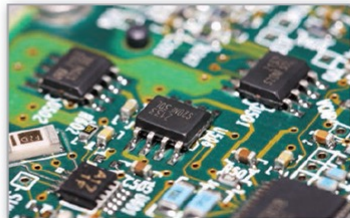
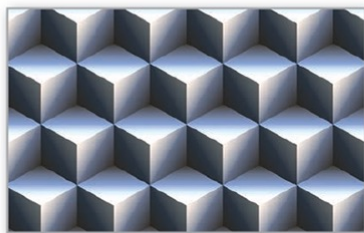
$$\begin{aligned}
 A' &= P^{-1}AP \\
 &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.
 \end{aligned}$$

Note that matrix A' is diagonal.

103

103

6 Linear Transformations



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104

6.5

Applications of Linear Transformations

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105

Objectives

- Identify linear transformations defined by reflections, expansions, contractions, or shears in R^2 .
- Use a linear transformation to rotate a figure in R^3 .

106

106

The Geometry of Linear Transformations in $\mathbb{R}^2$

107

107

The Geometry of Linear Transformations in $\mathbb{R}^2$ (1 of 2)

The first part of this section gives geometric interpretations of linear transformations represented by 2×2 elementary matrices.

Elementary Matrices for Linear Transformations in $\mathbb{R}^2$

Reflection in y -Axis

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reflection in x -Axis

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

108

108

The Geometry of Linear Transformations in R^2 (2 of 2)

Elementary Matrices for Linear Transformations in R^2

Reflection in Line $y = x$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Horizontal Expansion ($k > 1$)
or Contraction ($0 < k < 1$)

$$A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

Horizontal Shear

$$A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

Vertical Expansion ($k > 1$)
or Contraction ($0 < k < 1$)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

Vertical Shear

$$A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

109

109

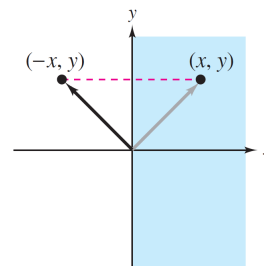
Example 1 – Reflections in R^2 (1 of 3)

The transformations below are **reflections**. These have the effect of mapping a point in the xy -plane to its “mirror image” with respect to one of the coordinate axes or the line $y = x$, as shown in Figure 6.5.

a. Reflection in the y -axis:

$$T(x, y) = (-x, y)$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$



Reflections in R^2

Figure 6.5(a)

110

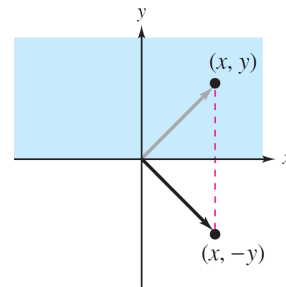
110

Example 1 – Reflections in R^2 (2 of 3)

b. Reflection in the x-axis:

$$T(x, y) = (x, -y)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$$



Reflections in R^2

Figure 6.5(b)

111

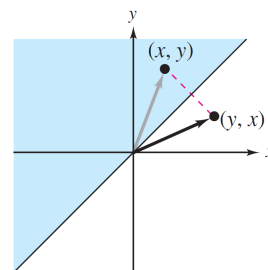
111

Example 1 – Reflections in R^2 (3 of 3)

c. Reflection in the line $y = x$:

$$T(x, y) = (y, x)$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$$



Reflections in R^2

Figure 6.5(c)

112

112

Example 2 – Expansions and Contractions in $\mathbb{R}^2$ (1 of 3)

The transformations below are **expansions** or **contractions**, depending on the value of the positive scalar k .

a. Horizontal expansions and contractions:

$$T(x, y) = (kx, y)$$

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} kx \\ y \end{bmatrix}$$

b. Vertical expansions and contractions:

$$T(x, y) = (x, ky)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ ky \end{bmatrix}$$

113

113

Example 2 – Expansions and Contractions in $\mathbb{R}^2$ (2 of 3)

Note in the figures below that the distance the point (x, y) moves by a contraction or an expansion is proportional to its x - or y -coordinate. For example, under the transformation represented by

$$T(x, y) = (2x, y)$$

the point $(1, 3)$ would move one unit to the right, but the point $(4, 3)$ would move four units to the right.

114

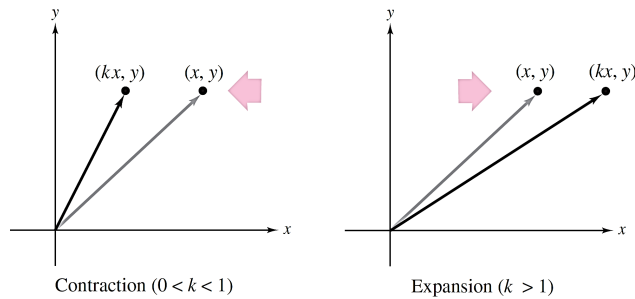
114

Example 2 – Expansions and Contractions in $\mathbb{R}^2$ (3 of 3)

Under the transformation represented by

$$T(x, y) = \left(x, \frac{1}{2}y\right)$$

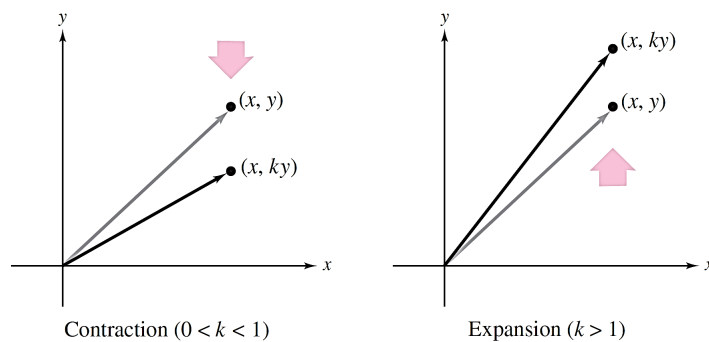
the point $(1, 4)$ would move two units down, but the point $(1, 2)$ would move one unit down.



115

115

Example 2 – Expansions and Contractions in $\mathbb{R}^2$ (4 of 3)



116

116

Example 3 – Shears in R^2 (1 of 3)

The transformations below are **shears**.

$$T(x, y) = (x + ky, y)$$

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ky \\ y \end{bmatrix}$$

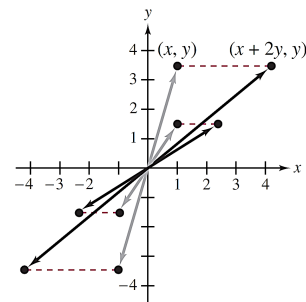
$$T(x, y) = (x, y + kx)$$

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ kx + y \end{bmatrix}$$

a. A horizontal shear represented by

$$T(x, y) = (x + 2y, y)$$

is shown at the right. Under this transformation, points in the upper half-plane “shear” to the right by amounts proportional to their y -coordinates.



117

117

Example 3 – Shears in R^2 (2 of 3)

Points in the lower half-plane “shear” to the left by amounts proportional to the absolute values of their y -coordinates. Points on the x -axis do not move by this transformation.

b. A vertical shear represented by

$$T(x, y) = (x, y + 2x)$$

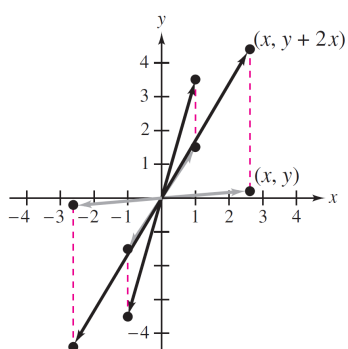
is shown below. Here, points in the right half-plane “shear” upward by amounts proportional to their x -coordinates.

118

118

Example 3 – Shears in R^2 (3 of 3)

Points in the left half-plane “shear” downward by amounts proportional to the absolute values of their x -coordinates. Points on the y -axis do not move.



119

119

Rotation in R^3

120

120

Rotation in R^3 (1 of 3)

Here you will see how linear transformations can be used to rotate figures in R^3 .

Say you want to rotate the point (x, y, z) counterclockwise about the z -axis through an angle θ , as shown in Figure 6.6

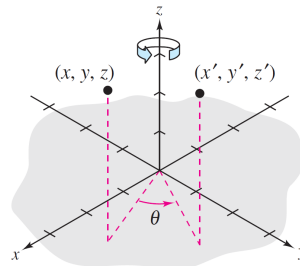


Figure 6.6

121

121

Rotation in R^3 (2 of 3)

Letting the coordinates of the rotated point be (x', y', z') , you have

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ z \end{bmatrix}.$$

122

122

Example 4 – Rotation About the z-Axis

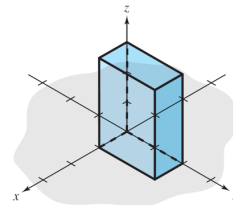
The eight vertices of the rectangular prism shown at the right are

$$V_1(0, 0, 0) \quad V_2(1, 0, 0)$$

$$V_3(1, 2, 0) \quad V_4(0, 2, 0)$$

$$V_5(0, 0, 3) \quad V_6(1, 0, 3)$$

$$V_7(1, 2, 3) \quad V_8(0, 2, 3).$$



Find the coordinates of the vertices after the prism is rotated counterclockwise about the z-axis through
(a) $\theta = 60^\circ$, (b) $\theta = 90^\circ$, and (c) $\theta = 120^\circ$.

123

123

Example 4 – Solution (1 of 4)

a. The matrix that yields a rotation of 60° is

$$A = \begin{bmatrix} \cos 60^\circ & -\sin 60^\circ & 0 \\ \sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying this matrix by the column vectors corresponding to each vertex produces the rotated vertices listed below.

$$V_1'(0, 0, 0) \quad V_2'(0.5, 0.87, 0) \quad V_3'(-1.23, 1.87, 0) \quad V_4'(-1.73, 1, 0)$$

$$V_5'(0, 0, 3) \quad V_6'(0.5, 0.87, 3) \quad V_7'(-1.23, 1.87, 3) \quad V_8'(-1.73, 1, 3)$$

124

124

Example 4 – Solution (2 of 4)

Figure 6.7(a) shows a graph of the rotated prism.

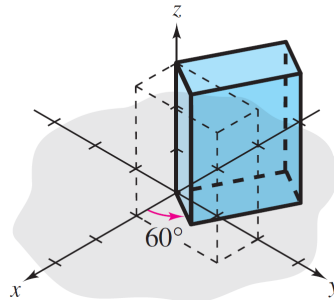


Figure 6.7(a)

125

125

Example 4 – Solution (3 of 4)

b. The matrix that yields a rotation of 90° is

$$A = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and Figure 6.7(b) shows a graph of the rotated prism.

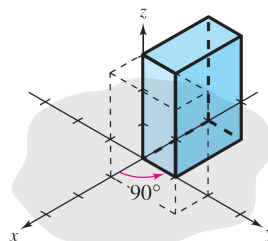


Figure 6.7(b)

126

126

Example 4 – Solution (4 of 4)

c. The matrix that yields a rotation of 120° is

$$A = \begin{bmatrix} \cos 120^\circ & -\sin 120^\circ & 0 \\ \sin 120^\circ & \cos 120^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and Figure 6.7(c) shows a graph of the rotated prism.

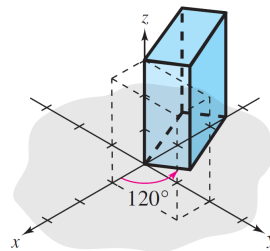


Figure 6.7(c)

127

127

Rotation in R^3 (3 of 3)

Example 4 uses matrices to perform rotations about the z-axis. Similarly, you can use matrices to rotate figures about the x- or y-axis.

Rotation About the x-Axis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

Rotation About the y-Axis

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

Rotation About the z-Axis

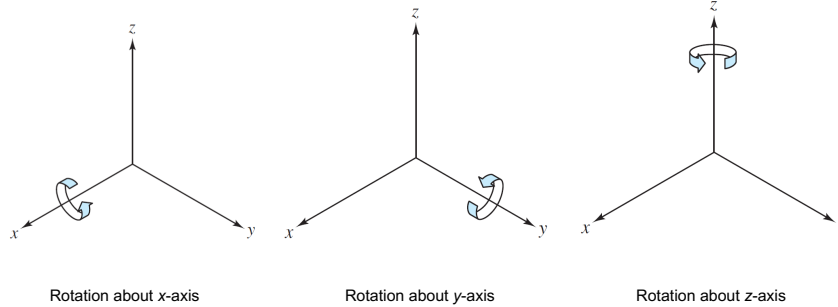
$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

128

128

Rotation in R^3 (4 of 3)

In each case, the rotation is oriented counterclockwise (using the “right-hand rule”) relative to the specified axis, as shown below.



129